Zap Lab

Electricity is energy that moves through a complete loop called a circuit. In Zap Lab you build, draw, measure, and fix small battery circuits, then show what you know at stations.

My name: _____

My partner: _____

Our meetings

Lesson	Date	What we'll do
Lesson 1	Fri Oct 16	Light It Up: What Is Zap Lab?
Lesson 2	Fri Nov 20	Switches, Series, and Parallel
Lesson 3	Fri Jan 15	Meters, LEDs, and Circuit Detectives
Lesson 4	Fri Feb 12	Mock Tournament
Lab I1	Before Lesson 2	Symbols and Schematics (home or an open Friday)
Lab I2	Before Lesson 3	Straw Resistance and Meter Scales (home or an open Friday)
Lab I3	Before Lesson 4	Safe Electricity and Flashlight Detective (home or an open Friday)

At the tournament

Pairs rotate through timed stations for up to 50 minutes (sorting materials, reading circuit drawings, looking at real low-voltage circuits and meters, answering questions) and bring only pencils, no notes.

Keep every handout for this event behind this divider, in order. Bring this binder to every Diagram & Decode Fridays meeting.

Zap Lab: What I can do

By tournament day, you should be able to do everything on this list. Check **Practicing** when you've started, and **Got it!** when you can do it without help. The lesson column tells you which handout to look back at: L = a group meeting, I = an independent lab you do at home or on an open Friday.

What I can do	Lesson	Practicing	Got it!
I can name sources of electricity (batteries, outlets, power plants, generators, solar panels) and say that electricity is energy that powers devices.	L1 I3	☐	☐
I can sort materials into conductors and insulators and explain the difference.	L1 L4	☐	☐
I can name the parts of a circuit: power source, wires, switch, load.	L1	☐	☐
I can explain open vs closed circuits and what a switch does.	L2 I1	☐	☐
I can draw and read symbols for cell, battery, wire, lamp, switch, resistor, LED, motor, buzzer, meter.	L2 I1	☐	☐
I can predict from a picture or schematic whether a bulb will light, and say why.	L2 L4 I1	☐	☐
I can build and compare series and parallel circuits.	L2 L4 I2	☐	☐
I can describe how current flows around a complete loop, from the battery, through the load, and back.	L2 I1	☐	☐
I can explain voltage as the push and match units: volts (V), amps (A), ohms (Ω), watts (W).	L3 L4 I2	☐	☐
I can explain that resistance slows current and that an LED (a diode) works only one way.	L3 I2	☐	☐
I can read a multimeter or meter scale and say what the number means.	L3 L4 I2	☐	☐
I can troubleshoot a broken circuit: gap, open switch, dead battery, burned-out bulb, backwards LED.	L3 L4 I3	☐	☐
I can name everyday uses of electricity and the energy each makes (light, heat, sound, motion).	I3	☐	☐
I can explain and follow electrical safety rules.	L1 L4 I3	☐	☐
I can answer station questions on diagrams, models, and real situations with no notes.	L4	☐	☐

Light It Up: What Is Zap Lab?

TODAY I CAN

- I can name sources of electricity (batteries, outlets, power plants, generators, solar panels) and say that electricity is energy that powers devices.
- I can sort materials into conductors and insulators and explain the difference.
- I can name the parts of a circuit: power source, wires, switch, load.
- I can explain and follow electrical safety rules.

Electricity is a kind of energy. It can make light, heat, sound, and motion. It only flows when it has a complete path, called a circuit.

Today's steps: run it yourselves (22 minutes)

You need: 4 AA batteries, 4 loose bulbs, 2 bulbs in holders, 2 battery holders with batteries, 10 clip leads, conductor test bag, sample station card. Set the timer for each step.

1. Event Judge 3 min. Set the timer for 3 minutes. Read these safety rules aloud: only use kit batteries; never touch a wall outlet; never clip a wire straight from + to - (it gets hot); if anything feels warm, let go and tell an adult. Each person says one rule back.
2. Both 7 min. Take 1 AA battery, 1 loose bulb, and 1 clip lead. Make the bulb light using only these three things. Stuck after 4 minutes? Find the two metal spots on the bulb and give the electricity a way back to the battery.
3. Partner A 2 min. Build a tester: battery holder to bulb holder with one clip lead. Clip one more lead to the other side of the bulb holder and one to the other battery wire. Touch the two loose clip ends together to check that the bulb lights.
4. Both 7 min. Fill in the Conductor tester table. For each object, write what it is made of and your prediction (C or I) first. Then touch both clip ends to it and record if the bulb lit. Test one object of your own in the last row.
5. Event Captain 3 min. Read the sample station card aloud; each pair writes its answer on the whiteboard. Then answer the Think about it questions.

Check your work: Ask your Event Captain for the Lesson 1 answer card. Check your Conductor tester table and Think about it answers. Fix any you missed in a different color.

Key words

Circuit	A complete loop that electricity flows around.
Circuit parts	Power source (gives energy), wires (the path), switch (opens or closes the path), load (uses the energy, like a bulb or motor).
Sources	Batteries, wall outlets (fed by power plants), generators, solar panels.
Conductor	A material that lets electricity flow easily. Most metals.
Insulator	A material that blocks electricity. Plastic, rubber, wood, glass, paper, cloth.

Conductor tester

Object	What is it made of?	My prediction (C or I)	Did the bulb light?
paper clip			
rubber band			
foil			
craft stick			
coin			
plastic spoon			

Think about it

1. Why did the bulb only light when you made a full loop?

2. A wire has metal inside and plastic outside. Why does it need both?

3. Name three sources of electrical energy.

At home before next time: Do Independent Lab I1 (Symbols and Schematics) before Lesson 2. Also make 6 flashcards (circuit, power source, load, switch, conductor, insulator) and quiz yourself twice. Case File: open Case 1, The Sneaky Conductors (p. 7), sort 12 things at home into conductors and insulators, and find the parts of a flashlight to earn the Conductor Detective badge.

Switches, Series, and Parallel

TODAY I CAN

- I can explain open vs closed circuits and what a switch does.
- I can draw and read symbols for cell, battery, wire, lamp, switch, resistor, LED, motor, buzzer, meter.
- I can predict from a picture or schematic whether a bulb will light, and say why.
- I can build and compare series and parallel circuits.
- I can describe how current flows around a complete loop, from the battery, through the load, and back.

A switch opens and closes a circuit. When you add a second bulb, you can hook it up in series (one path) or in parallel (two or more paths, called branches).

Today's steps: run it yourselves (22 minutes)

You need: 2 battery holders with fresh batteries, 4 bulbs in holders, 2 switches, 12 clip leads, 'Will it light?' plates A-D. Set the timer for each step.

1. Both 4 min. Set the timer for 4 minutes. Connect the battery holder, switch, and one bulb in a loop. Close the switch, then open it. Trace the path with a finger from + around to -. Answer Check yourself 1 and 2.
2. Both 5 min. Will it light? For each plate A-D, write your prediction first. Then lift the sticky note and check. For every no, write where the problem is.
3. Both 6 min. Read the Series line in the Remember box. Connect battery, bulb 1, bulb 2, and back to the battery in one loop. Fill in the Series row: compare to one bulb alone, then unscrew bulb 1 and see what bulb 2 does.
4. Both 7 min. Read the Parallel line. Give each bulb its own branch: each bulb holder gets one lead to the battery's + wire and one to the - wire. Fill in the Parallel row the same way. Unhook the battery, then answer Check yourself 3 and 4.

Check your work: Ask your Event Captain for the Lesson 2 answer card. Check your plate table, Test results, and Check yourself answers. Fix any you missed in a different color.

Will it light? (plates A-D)

Plate	My prediction (yes/no)	Result	If no, where is the problem?
A			
B			
C			
D			

Test results

	Brightness (brighter, same, or dimmer than 1 bulb)	Unscrew one bulb. What does the other do?
Series		
Parallel		

Remember

Series	One path. Current goes through every part in order. One break stops everything. Bulbs share the push, so they are dimmer.
Parallel	Two or more paths. Each branch works on its own. Bulbs stay bright. Homes and schools are wired this way.

Check yourself

1. Switch open: is the circuit open or closed, and is the bulb on or off? Switch closed?

2. Describe the path the current takes around your circuit.

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3. Which kind of circuit is used for the lights in your classroom? Why?

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4. Two bulbs are in parallel. One burns out. What happens to the other?

At home before next time: Do Independent Lab I2 (Straw Resistance and Meter Scales) before Lesson 3. Also make 2 flashcards, series and parallel, with a small schematic on the back of each, and quiz yourself on all your cards. Case File: open Case 2, The Dark Bulbs (p. 8), and crack all 8 Will it light? drawings to earn the Loop Master badge.

Meters, LEDs, and Circuit Detectives

TODAY I CAN

- I can explain voltage as the push and match units: volts (V), amps (A), ohms (Ω), watts (W).
- I can explain that resistance slows current and that an LED (a diode) works only one way.
- I can read a multimeter or meter scale and say what the number means.
- I can troubleshoot a broken circuit: gap, open switch, dead battery, burned-out bulb, backwards LED.

Today you measure real batteries, control an LED with resistors, and fix broken circuits like a detective. Check one thing at a time.

Today's steps: run it yourselves (27 minutes)

You need: 2 multimeters (already set to DC 20 V), 4 fresh AA and 2 used AA batteries (taped), 2 nine-volt batteries, 2 battery holders with batteries, 4 red LEDs, resistor card (100, 1k, 10k ohms), 8 clip leads, 6 troubleshooting bags A-F. Set the timer for each step.

1. Partner A 7 min. Set the timer for 7 minutes. Do not turn the meter dial. Touch red to + and black to - and read each battery while Partner B records in the Meter readings table. Swap the probes and read again (a minus sign means the probes are backwards). Switch jobs halfway.
2. Both 8 min. Build: battery holder, 100 ohm resistor, LED (long leg toward +), and back. Swap in the 1k, then the 10k resistor, and fill in the LED and resistors table. Flip the LED around for the last row. Always keep a resistor in with the LED.
3. Both 12 min. Circuit detectives: the Event Judge sets the timer for 2 minutes per bag and calls 'Rotate!' Use the Detective checklist, one thing at a time, and write the problem and fix in the table. After you write, peek at the answer strip folded under the bag label. Answer Check yourself if time is left.

Check your work: Ask your Event Captain for the Lesson 3 answer card. Check your meter readings, LED table, detective table, and Check yourself answers. Fix any you missed in a different color.

Meter readings

Battery	Reading (V)	Probes swapped (V)
fresh AA		
used AA		
2-AA holder		
9 V		

LED and resistors

Resistor in the circuit	LED brightness (bright, dim, very dim, off)
100 ohms	
1,000 ohms (1k)	
10,000 ohms (10k)	
100 ohms, LED flipped around	

Detective checklist

Check one thing at a time: (1) Is the battery good? Use the meter. (2) Is every clip on metal? (3) Is the loop complete? (4) Is the switch closed? (5) Is the bulb burned out, or the LED backwards? (6) Is an insulator in the path?

Circuit detectives

Bag	What was wrong?	How we fixed it
A		
B		
C		
D		
E		
F		

Check yourself

1. One AA reads 1.6 V. Another reads 1.1 V. Which is running low? How do you know?

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2. What happened to the LED as you added more ohms? Why?

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3. Why did the LED stay dark when you flipped it around?

At home before next time: Do Independent Lab I3 (Safe Electricity and Flashlight Detective) before Lesson 4, the mock tournament. Then shuffle ALL your Zap Lab flashcards and quiz yourself with no notes; go through any missed cards 3 more times. Case File: open Case 3, The Mystery Meter (p. 9), read the 4 meters and find 4 volt or watt labels at home to earn the Meter Reader badge.

Mock Tournament

TODAY I CAN

- I can sort materials into conductors and insulators and explain the difference.
- I can predict from a picture or schematic whether a bulb will light, and say why.
- I can build and compare series and parallel circuits.
- I can explain voltage as the push and match units: volts (V), amps (A), ohms (Ω), watts (W).
- I can read a multimeter or meter scale and say what the number means.
- I can troubleshoot a broken circuit: gap, open switch, dead battery, burned-out bulb, backwards LED.
- I can explain and follow electrical safety rules.
- I can answer station questions on diagrams, models, and real situations with no notes.

Mock tournament answer sheet. No notes. Work with your partner. When time is called, stop and move.

Today's steps: run it yourselves (27 minutes)

You need: 5 station cards A-E, Station A: bag of 8 objects, Station B: battery holder, 2 bulbs in holders, switch, 6 clip leads, Station D: meter set to DC 20 V and a taped AA, Station E: fault bag. Set the timer for each step.

1. Event Judge 1 min. Read the rules aloud: stay at your station until time is called, no going back, no notes, whisper with your partner only. One pair starts at Station A, the other at B.
2. Event Judge 4 min. Set the timer for 4 minutes. Pairs work their first station, then the Judge calls 'Rotate!'
3. Both 4 min. Second station. Move forward one letter when time is called.
4. Both 4 min. Third station.
5. Both 4 min. Fourth station.
6. Both 4 min. Fifth station. Skip any question you are stuck on and come back if time is left.
7. Event Captain 6 min. Read the answers from the answer card while each pair marks its own sheet. Each pair circles its 2 weakest topics and writes them in the at-home box.

Check your work: Your Event Captain reads the Lesson 4 answer card aloud. Mark your own sheet and fix any you missed in a different color. Your 2 weakest topics are your study list.

Station A: Conductors and insulators

1. List the conductors, then the insulators.

2. What do most conductors have in common?

Station B: Build it

1. Build: battery, then a switch, then two lamps in parallel. Raise your hand when both light. If you open the switch, what happens to both lamps? Why?

Station C: Will it light? Series or parallel?

Circuit	Your answer
1. Battery, bulb, and an open switch in one loop (lights or not?)	
2. Both bulb wires go to the + end of the battery (lights or not?)	
3. Battery, wire, bulb, wire back to the battery (lights or not?)	
4. One path through three bulbs (series or parallel?)	
5. Each bulb has its own branch (series or parallel?)	

Station D: Meters and units

1. Measure the AA battery. Reading: _____ V

2. Write the unit for voltage ___ current ___ resistance ___ power ___

3. Four 1.5 V batteries are in series. Total volts?

Station E: Troubleshooting and safety

1. The LED in the bag will not light. What is wrong? How do you fix it?

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2. Is it safe to use a radio plugged in next to a sink full of water? Why?

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3. A power line is down in your yard after a storm. What should you do?

At home before next time: Final review before the tournament (20 min). Write your 2 weakest topics from today here: ___ and ___. Quiz yourself on those flashcards until you get them right 3 times. Then have a family member quiz you on 10 random cards with no peeking. Get a good night's sleep before tournament day. Case File: open Case 4, The Secret Blueprints (p. 10), and do Describe & Draw with your partner or a family member to earn the Describe & Draw badge.

Lab I1: Symbols and Schematics

THIS LAB HELPS ME

- I can explain open vs closed circuits and what a switch does.
- I can draw and read symbols for cell, battery, wire, lamp, switch, resistor, LED, motor, buzzer, meter.
- I can predict from a picture or schematic whether a bulb will light, and say why.
- I can describe how current flows around a complete loop, from the battery, through the load, and back.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. Scientists draw circuits with simple symbols instead of pictures. The drawing is called a schematic (skee-MAT-ik). At the tournament you will read schematics with no notes, so learn these symbols by heart.

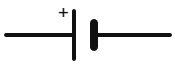
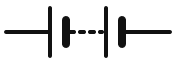
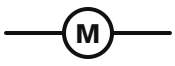
Steps (28 minutes)

You need: this handout, 10 index cards, a colored pencil, a flashlight (at home, optional). Set the timer for each step.

1. Me 5 min. Set a timer for 5 minutes. Read the Symbol chart aloud. Copy each symbol into the Draw it column.
2. Me 6 min. Make 10 flashcards: draw a symbol on the front and write its name on the back (cell, battery, wire, lamp, switch, resistor, LED, motor, buzzer, meter).
3. Me 4 min. Quiz yourself: look at each symbol and say its name. Keep going until you get all 10 right twice in a row. If your partner is with you, quiz each other.
4. Me 5 min. Fill in the Will it light? table. For each drawing, write yes or no, and if no, write where the problem is. Remember: electricity needs a full loop from + back to -.
5. Me 5 min. In the Draw it box, draw a battery, a closed switch, and a lamp in one loop. Draw arrows with a colored pencil to show the current going from + around to -. Then answer the Check yourself questions.
6. Me 3 min. Flashlight schematic: look at a flashlight at home (or read the description in the box). Draw it as a schematic with 2 cells, a switch, and a lamp.

Check your work: At home, check your drawings against the Symbol chart. On an open Friday or at Lesson 2, ask for the Lab I1 answer card from the Zap Lab bin, check your Will it light? table and questions, and fix any you missed in a different color.

Symbol chart

Part	Symbol	What it looks like	Draw it
Cell		One long line (+) and one short line (-) side by side. One AA battery is one cell.	
Battery		Two or more cells joined together, drawn as long-short-long-short lines.	
Wire		A straight line. Wires meet with a dot where they join.	
Lamp (bulb)		A circle with an X inside.	
Switch (open)		A line with a gap and a short line tilted up, like a gate.	
Resistor		A zigzag line (or a small rectangle).	
LED (diode)		A triangle pointing at a line, with two small arrows coming out (light).	
Motor		A circle with an M inside.	

Buzzer		A half circle (dome) on a line.	
Meter		A circle with V (voltmeter) or A (ammeter) inside.	

Will it light?

Drawing	Lights? (yes/no)	If no, where is the problem?
1. Wire from the + end to the lamp, and a second wire from the lamp back to the - end.		
2. Both lamp wires go to the + end of the battery.		
3. Battery, lamp, and a switch drawn open (the gate is tilted up) in one loop.		
4. The same loop with the switch closed.		
5. Wire from + to the lamp. The lamp's other wire stops short and does not touch the battery.		
6. One wire goes straight from + to -. The lamp is on a side wire that touches only the + end.		

Draw it: battery, closed switch, and lamp in one loop (add current arrows)

Flashlight schematic

A flashlight has 2 cells end to end, a switch, and a lamp in one loop.

Check yourself

1. What does a switch do?

2. The switch is open. Is the circuit open or closed? Is the lamp on or off?

3. Which way does current flow around a circuit?

Bring to the next meeting: Bring this handout with your drawings and your 10 symbol flashcards.

Lab I2: Straw Resistance and Meter Scales

THIS LAB HELPS ME

- I can build and compare series and parallel circuits.
- I can explain voltage as the push and match units: volts (V), amps (A), ohms (Ω), watts (W).
- I can explain that resistance slows current and that an LED (a diode) works only one way.
- I can read a multimeter or meter scale and say what the number means.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. You cannot see electricity move, so scientists use models. In this lab your breath is the push (voltage), the air flowing is the current, and the straws are resistance. Then you practice reading meter scales like at the tournament.

Steps (30 minutes)

You need: 3 straws and 1 coffee stirrer, tape, a timer, 6 index cards, this handout. Set the timer for each step.

1. Me 5 min. Read the Words and units box aloud. Make 6 flashcards: voltage (V), current (A), resistance (ohms), power (W), resistor, diode. Word on the front, meaning and unit on the back.
2. Me 8 min. Straw test. Take one big breath and blow out gently through 1 straw while the timer runs; stop the timer when you are out of air. Do each setup in the Straw data table twice. Tape 2 straws end to end for series; hold 2 straws side by side for parallel. Rest between tries, and stop if you feel dizzy.
3. Me 5 min. Answer the Think like an electrician questions. Then draw both schematics in the box: 2 lamps in series and 2 lamps in parallel.
4. Me 6 min. Answer the Read the scale questions. For a scale: find the numbers on each side of the needle, count the small marks, and figure out what each mark is worth.
5. Me 3 min. Battery label hunt: find 3 batteries (look only) and fill in the table with the size and the volts printed on each.
6. Me 3 min. Quiz yourself on your 6 new flashcards until you get them all right twice. If your partner is here, quiz each other.

Check your work: Check your straw results against the Words and units box: more resistance should mean slower flow. On an open Friday or at Lesson 3, ask for the Lab I2 answer card from the Zap Lab bin and fix any you missed in a different color.

Words and units

Voltage (V, volts)	The push that gives current energy to move. One AA = 1.5 V.
Current (A, amps)	How much electricity flows past a point.
Resistance (Ω, ohms)	How much something slows current. More ohms = less current.
Power (W, watts)	How fast a device uses energy.
Resistor	A part that adds resistance on purpose, to protect other parts.
Diode / LED	Lets current flow one way only. LED = light-emitting diode. Long leg = +.

Straw data (seconds to blow out one big breath)

Setup	Try 1 (sec)	Try 2 (sec)	Easier or harder than 1 straw?
1 straw			(this is your normal)
2 straws taped end to end (like series)			
2 straws side by side (like parallel)			
1 thin coffee stirrer			

Think like an electrician

1. Which setup took the longest to empty your lungs? What does that tell you about its resistance?

_____

2. More resistance means _____ current. Which bulb circuit is like the 2 straws end to end?

3. Why was it easier with 2 straws side by side?

4. An LED is a diode, like a one-way door. What happens if you put it in backwards?

Draw both schematics (battery + 2 lamps each)

Left half: series. Right half: parallel.

Read the scale

1. A scale is marked 0, 5, 10 V. Between 0 and 5 there are 9 small marks (each = 0.5 V). The needle is on the 5th small mark. Reading?

2. Between 5 and 10 V there are 4 small marks (each = 1 V). The needle is on the 2nd mark after 5. Reading?

3. One AA reads 1.6 V. Another reads 1.1 V. Which is running low? How do you know?

4. Two AA batteries (1.5 V each) are in series. About how many volts?

5. An LED and resistor are hooked to a battery, but the LED is dark. What could be wrong?

Battery label hunt

Where I found it	Size (AA, AAA, 9 V, coin…)	Volts printed on it

Bring to the next meeting: Bring this handout with your straw data and your 6 unit flashcards.

Lab I3: Safe Electricity and Flashlight Detective

THIS LAB HELPS ME

- I can name sources of electricity (batteries, outlets, power plants, generators, solar panels) and say that electricity is energy that powers devices.
- I can troubleshoot a broken circuit: gap, open switch, dead battery, burned-out bulb, backwards LED.
- I can name everyday uses of electricity and the energy each makes (light, heat, sound, motion).
- I can explain and follow electrical safety rules.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. Electricity from an outlet is much stronger than a battery and can hurt you, so every Zap Lab scientist knows the safety rules. You will also hunt for electricity around you and be a detective with a flashlight.

Steps (28 minutes)

You need: this handout, a flashlight (at home), open Friday instead: battery holder, bulb in holder, 3 clip leads, paper slip from the Zap Lab bin. Set the timer for each step.

1. Me 5 min. Set a timer for 5 minutes. Read the Safety rules and Where outlet electricity comes from boxes aloud.
2. Me 7 min. Fill in the Safe or unsafe? table. For each unsafe one, write why. Hint: water and people are conductors, so electricity can pass through them.
3. Me 6 min. Electricity hunt: walk around your home (or the school room) and fill in 8 rows. Try to find at least one device that makes light, heat, sound, and motion. Look only; do not touch outlets or cords.
4. Me 7 min. Flashlight detective: do each test in the table and write what happened and why. Put the batteries back the right way when you finish. On an open Friday, build battery, bulb, and clip leads in a loop and try: all clipped, one clip undone, and a paper slip inside one clip.
5. Me 3 min. Answer the Check yourself questions.

Check your work: Read your Safe or unsafe answers against the Safety rules box. On an open Friday or at Lesson 4, ask for the Lab I3 answer card from the Zap Lab bin and fix any you missed in a different color.

Safety rules

Outlets and water	Only plugs go in outlets. Keep electric things away from water; dry your hands first. Water and people are conductors.
Cords	Pull the plug, not the cord. Tell an adult about damaged cords.
Power lines	Stay far away. Never touch a fallen line. Fly kites in open fields.
Short circuit	A wire straight from + to - with no load makes a battery hot.

Where outlet electricity comes from

Power plants	Big buildings where generators spin to make electricity.
Generators	Machines that turn motion into electricity. Steam, water, or wind can spin them.
Solar panels and wind turbines	Make electricity from sunlight and wind.
Power lines	Carry electricity from the plant to homes, schools, and outlets.
Batteries	Store energy in chemicals and give a small, safe push.

Safe or unsafe?

Situation	Safe or unsafe?	If unsafe, why?
1. Flying a kite near power lines		
2. Using a hair dryer next to a full bathtub		
3. Pulling a plug out by its cord		

Situation	Safe or unsafe?	If unsafe, why?
4. Using a cord with bare wire showing		
5. Putting a fork in a toaster		
6. Plugging many things into one outlet strip		
7. Using only the batteries from our kit		
8. Telling an adult when a battery gets warm		
9. Staying far from a downed power line		
10. Drying your hands before touching a switch		
11. Covering unused outlets around little kids		
12. Touching a 9 V battery to your tongue		

Electricity hunt

Device	Home, school, or community?	Source (battery or outlet)	Energy made (light, heat, sound, motion)

Flashlight detective

Test	What happened?	Why?
1. Turn the switch on.		
2. Turn the switch off.		
3. Take out one battery and turn the switch on.		
4. Put that battery back in the wrong way (flipped) and turn it on.		
5. Put all batteries back the right way and turn it on.		

Check yourself

1. Name three sources of electricity.

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2. Your flashlight will not turn on. Name two things to check, one at a time.

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3. Why should you never use a radio or hair dryer near water?

Bring to the next meeting: Bring this handout with your Safe or unsafe table, electricity hunt, and flashlight detective table filled in.